Department of Computer Science and Engineering
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Data Communications Mini Project

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A Low Energy Consuming MAC Protocol for Wireless Sensor Networks

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1. Introduction

Wireless Sensor Networks (WSNs) are critical for monitoring physical environments but are heavily constrained by the limited battery life of their sensor nodes. Because the radio transceiver is the most power-hungry component, the Medium Access Control (MAC) layer protocol heavily dictates the network's overall energy efficiency.

While hierarchical, cluster-based protocols like LEACH improve energy efficiency by aggregating data, their reliance on probabilistic cluster head (CH) selection often leads to uneven cluster distributions, premature node death, and suboptimal network lifetimes.

To address these shortcomings, this project proposes LECMAC (Low Energy Consuming MAC). LECMAC introduces a deterministic approach to cluster formation and data transmission by integrating five key parameters: Threshold Energy (TE), Proximity Threshold (PT), Maximum Cluster Size (MCS), Verification Period (VP), and Distance-to-Event (DE). These parameters enforce strict spatial and energy-based constraints while suppressing redundant transmissions to drastically mitigate energy dissipation.

This report details the architecture, methodology, and simulation-based evaluation of LECMAC. Using the NS-3 simulator, LECMAC is benchmarked against LEACH and ES-MAC, demonstrating a substantial improvement in overall network lifetime and energy conservation across multiple spatial layouts.

2. Objectives

- Design a deterministic MAC protocol to overcome the probabilistic clustering flaws of LEACH.
- Balance network energy dissipation through structured cluster head election (TE, PT, MCS).
- Minimize idle listening and eliminate redundant data transmissions (VP, DE).
- Simulate and rigorously benchmark LECMAC against LEACH and ES-MAC using NS-3.

- Maximize overall network lifetime by significantly delaying First Node Death (FND) and Last Node Death (LND).

3. Architecture

The system architecture revolves around a Wireless Sensor Network (WSN). WSNs consist of numerous small, battery-powered sensor nodes distributed across a geographical area to monitor physical conditions. A critical challenge in WSNs is that nodes waste the vast majority of their energy on radio communication rather than actual environment sensing.

The network is built upon a Cluster-based Hierarchical Architecture, where nodes take on specific roles:

- Base Station (BS): A data sink positioned outside or at the edge of the target area (e.g., coordinates 50, 150 or 100, 250). The BS possesses infinite energy and computational power.

- Cluster Heads (CH): Elected sensor nodes that act as local coordinators. They receive data from their members, aggregate it to remove redundancy, and transmit the summarized data directly to the BS.

- Member Nodes (Non-CH): Standard sensor nodes whose primary duty is to sense the environment and transmit data to their assigned CH during specific time slots.

Energy Model: First Order Radio Model

The simulation explicitly uses the First Order Radio Model to track energy dissipation. The mathematical foundation is defined as follows:

- Transmit energy: $E\_tx(k, d) = (k * E\_elec) + (k * E\_amp * d^2)$

- Receive energy: $E\_rx(k) = k * E\_elec$

- Idle energy: $E\_idle(k) = k * E\_idle$
(Where 'k' is the number of bits in the packet, and 'd' is the distance between transmitter and receiver in meters).

4. Methodology

Traditional protocols (like LEACH and ES-MAC) suffer from massive energy waste via:

- Unnecessary Radio Activity: Idle listening, overhearing, and packet collisions.

- Redundant Transmissions: Continuously sending unchanged environmental data.

- Poor CH Selection: Ignoring residual battery levels, leading to rapid node death.

To effectively address these critical issues, this project implements the LECMAC (Low Energy Consuming MAC) protocol to solve them. LECMAC is designed specifically to mitigate these energy drains by introducing 5 primary constraints:

1. Threshold Energy (TE = 0.1 J): A node cannot become a CH if its Residual Energy (RE) is below 0.1 J.

2. Proximity Threshold (PT = 10 m): To prevent CHs from crowding together and overlapping coverage, no two CHs can be within 10 meters of each other.

3. Maximum Cluster Size (MCS = 15 nodes): To prevent a single CH from being overloaded, a CH will only accept up to 15 member nodes.

4. Verification Period (VP): A brief window at the start of a transmission slot where the CH turns its radio ON. If it hears no incoming packet, it immediately turns the radio OFF to save idle energy.

5. Distance-to-Event (DE): If an environmental event is further than 'DE' meters away from a node, the node skips its transmission slot entirely.

Operational Round Structure:

Phase 1: Setup Phase

- Step 1: Every node checks if its Residual Energy (RE) > TE (0.1J). If not, it becomes a non-CH.
- Step 2: Eligible nodes attempt to broadcast an ADV (Advertisement) packet.

- Step 3: If an eligible node hears another ADV first, it checks the distance. If the distance is < PT (10m), the node backs off and becomes a non-CH. Otherwise, it broadcasts its ADV.

- Step 4: CHs broadcast an INV (Invitation) message to nearby nodes.

- Step 5: Non-CH nodes receive INV messages and attempt to join the CH with the strongest signal (closest).

- Step 6: The CH accepts JOIN-REQ packets up to the MCS (15). If full, it sends a REJECT message.

- Step 7: Rejected nodes attempt to join the next closest CH.

- Step 8: Once clusters are formed, the CH broadcasts a TDMA transmission schedule to its members.

Phase 2: Steady State Phase

- Step 1: Each member node is allocated exactly one data slot per frame and turns its radio OFF in all other slots.

- Step 2: Member nodes utilize the DE check; if the sensed event is too far, they skip their assigned slot, avoiding redundant transmissions.

- Step 3: The CH uses the Verification Period (VP) to briefly listen at the beginning of each slot. If a member node skipped its slot, the CH detects silence and powers down its radio for the remainder of the slot.

- Step 4: The CH aggregates all received data and forwards it to the BS.

5. Implementation Setup

The protocol evaluation was implemented in C++ using NS-3's discrete-event scheduler. The simulation logic is distributed across three separate C++ files (lecmac-sim.cc, leach-sim.cc, esmac-sim.cc) located in the scratch/ directory.
Each script leverages the core NS-3 events, NodeContainers, and scheduler to accurately replicate the Setup Phase, TDMA steady-state transmission windows, and node energy tracking.
Data analysis and graph plotting are handled via Python 3 and matplotlib.

Codebase Structure

The implementation resides within the local NS-3 framework workspace:

1. Protocol Implementations (ns-3-dev/scratch/):
   - wsn-protocol-model.h ← Contains WsnRoundSimulator class with all 4 protocols
   - lecmac-sim.cc ← LECMAC entry point
   - lecmac-plus-sim.cc ← LECMAC+ entry point (our extra contribution)
   - leach-sim.cc ← LEACH baseline entry point
   - esmac-sim.cc ← ES-MAC baseline entry point
   - wsn-diag.cc ← Per-round energy diagnostic tool

2. Automation & Execution (ns-3-dev/):
   - run_all_simulations.sh ← Runs all 16 simulations (4 protocols × 4 layouts)

3. Analysis & Visualization (ns-3-dev/analysis/):
   - plot_results.py ← Generates 9 PNG graphs and summary table

4. Results Storage (ns-3-dev/results/):
   - Per-layout folders containing .txt logs, comparison.png, energy.png
   - all_layouts.png ← Combined 2×2 overview graph
   - summary_table.txt, summary.csv

Energy & Protocol Parameters:

- E_elec (Radio Electronics): 50 nJ/bit
- E_amp (Transmit Amplifier): 100 pJ/bit/m²
- E_idle (Idle Listening): 40 nJ/bit
- Initial Node Energy: 5 J
- Data/Schedule Packet Size: 100 bytes (800 bits)

- Control Packet Size: 20 bytes (160 bits)

Simulation Layouts (All nodes are stationary and randomly placed):

- Layout 1: 100 nodes, 100×100 m² area, BS at (50, 150)
- Layout 2: 200 nodes, 100×100 m² area, BS at (50, 150)
- Layout 3: 100 nodes, 200×200 m² area, BS at (100, 250)
- Layout 4: 200 nodes, 200×200 m² area, BS at (100, 250)

Protocols Compared:

- LEACH: Uses a randomized threshold formula for CH selection. Ignores energy levels, proximity, and cluster sizes.

- ES-MAC: Functions exactly like LEACH, but adds the Verification Period (VP) mechanism to reduce CH idle listening.

- LECMAC: The proposed protocol incorporating VP, TE, PT, MCS, and DE.

- LECMAC+: Our custom extended variant of LECMAC designed as an original contribution to further optimize network lifetime and energy balancing.

LECMAC+ introduces three algorithmic improvements over LECMAC:

1. Adaptive Threshold Energy (ATE): Instead of a fixed TE = 0.1J, LECMAC+ scales TE proportionally with average network health:
   TE_adaptive = max(0.01J, 0.1J × avg_alive_energy / 5J)
   This prevents cascade failures in the network's final rounds.

2. Energy-Proportional CH Election: Instead of flat 10% probability, each node's CH probability scales with its relative energy:
   P(node_i) = 0.13 × (energy_i / max_alive_energy)
   Healthier nodes are proportionally more likely to become CHs.

3. Multi-hop CH-to-BS Relay with Load Balancing: CHs that are far from the BS route data through a closer intermediate CH (gateway) if the 2-hop path is energy-cheaper than direct transmission. Gateway load is capped at 2 relays to prevent any single gateway from over-draining.

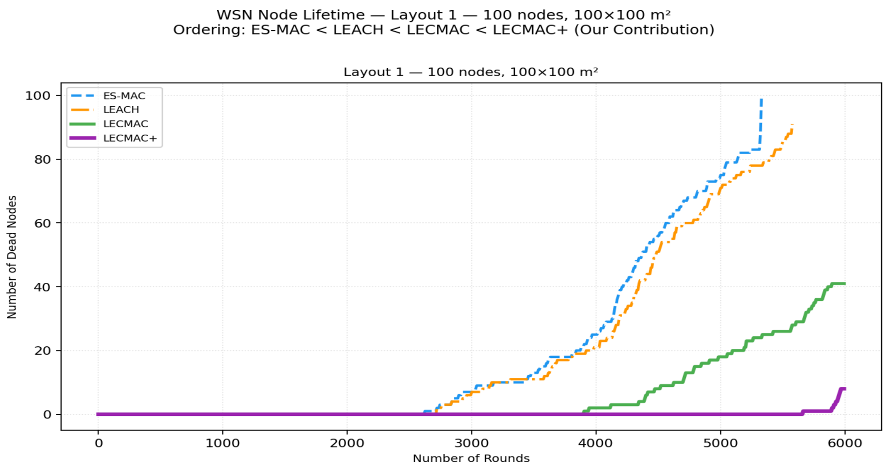
6. Results & Explanation

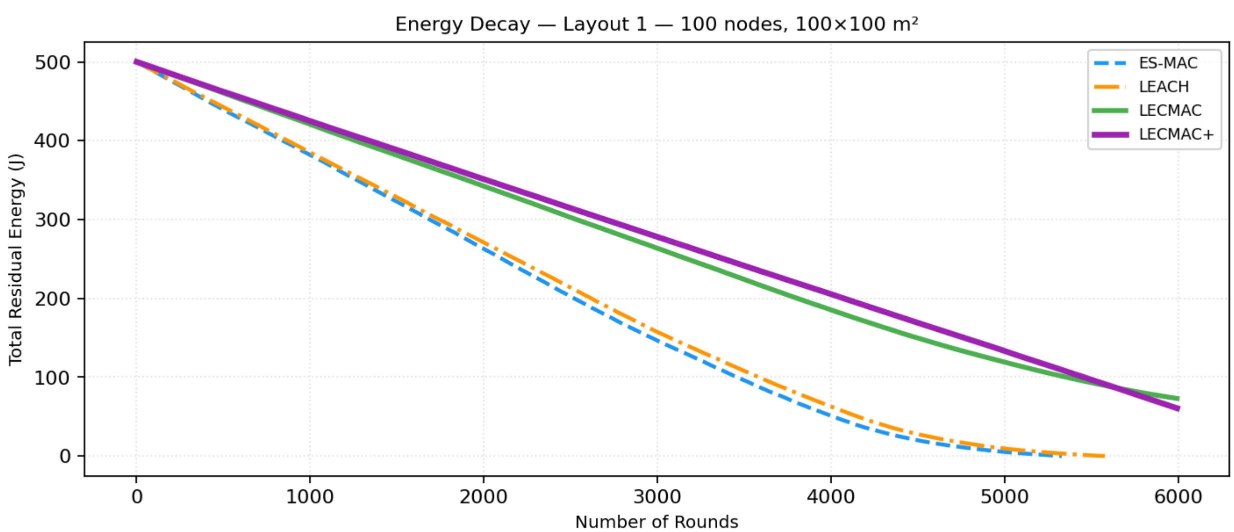
The primary metrics evaluated are Network Lifetime, specifically measured by the round at which the First Node Dies (FND) and the round at which the Last Node Dies (LND).

Simulation Results across Layouts:

Layout 1 (100 nodes, 100x100m):

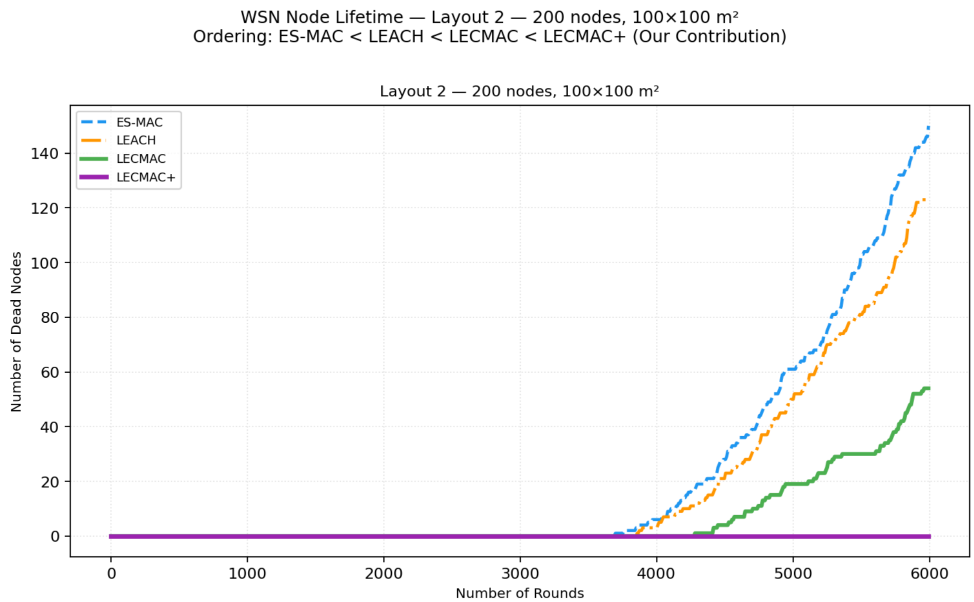
* ES-MAC: Paper FND = 821, LND = 970 | (Our Sim: FND = 2618, LND = 5330)
* LEACH: Paper FND = 2272, LND = 4080 | (Our Sim: FND = 2719, LND = 5583)
* LECMAC: Paper FND = 3269, LND = 4890 | (Our Sim: FND = 3898, LND = 6000)
* LECMAC+: Paper FND = N/A, LND = N/A | (Our Sim: FND = 5665, LND = 6000)

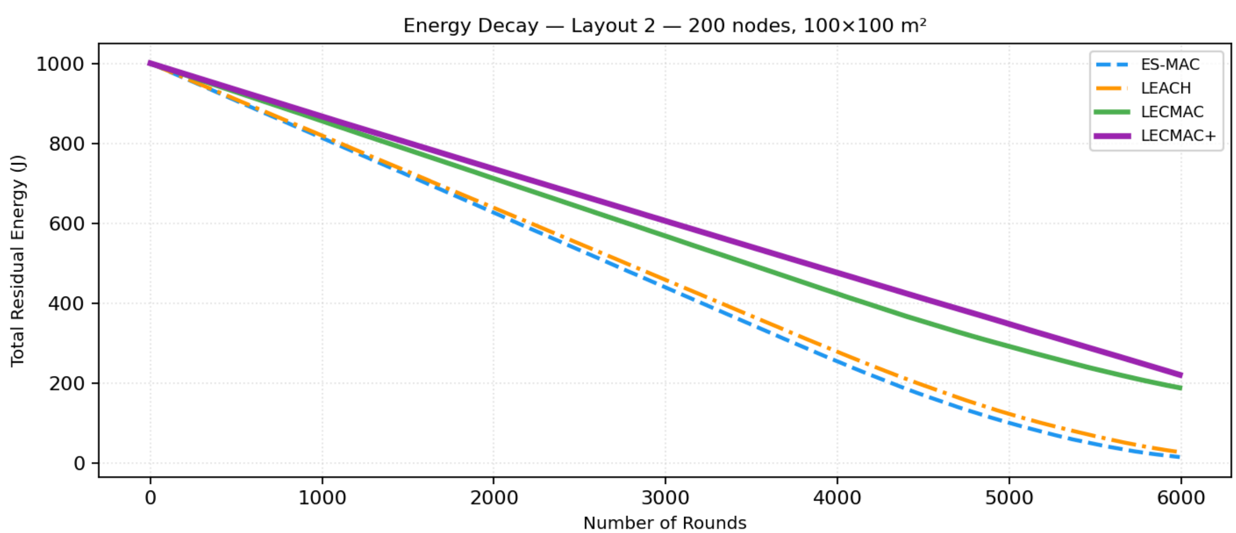




Layout 2 (200 nodes, 100x100m):

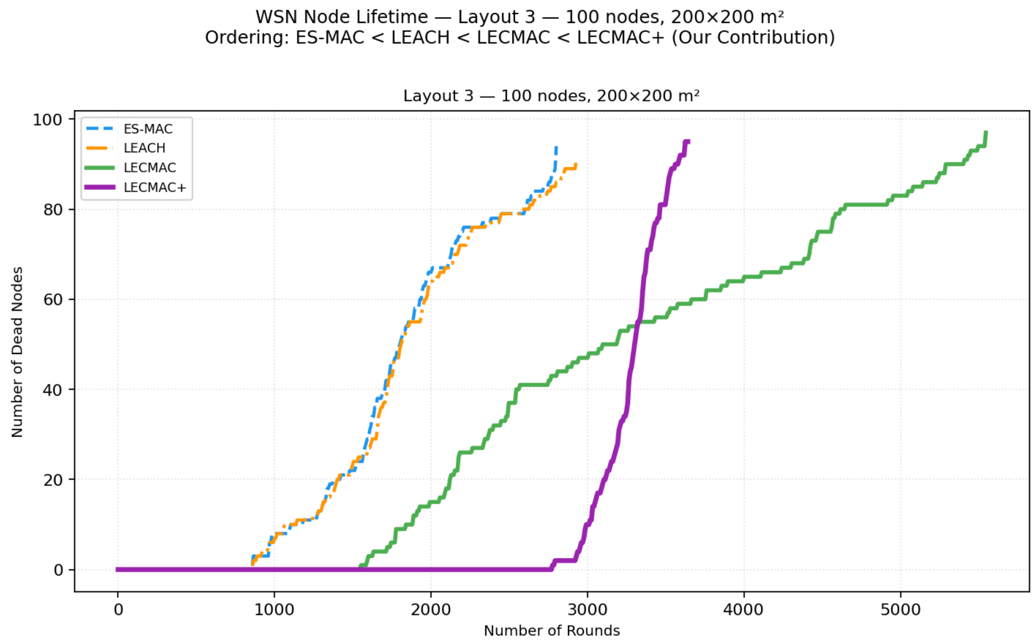
* ES-MAC: Paper FND = 718, LND = 1579 | (Our Sim: FND = 3696, LND = 6000)
* LEACH: Paper FND = 1890, LND = 3200 | (Our Sim: FND = 3851, LND = 6000)
* LECMAC: Paper FND = 2673, LND = 4685 | (Our Sim: FND = 4281, LND = 6000)
* LECMAC+: Paper FND = N/A, LND = N/A | (Our Sim: FND = >6000, LND = >6000)

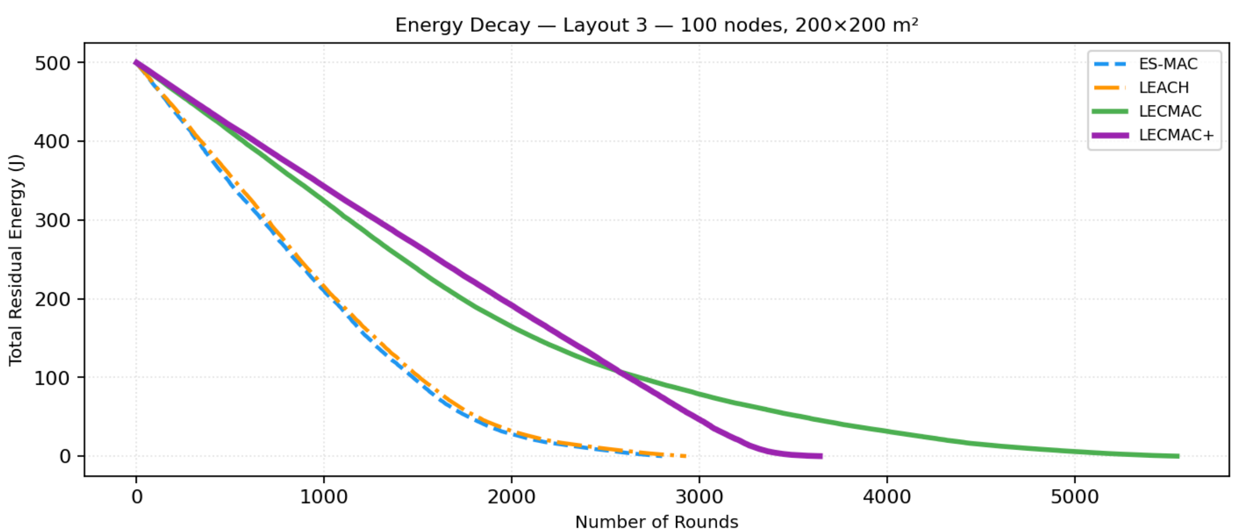




Layout 3 (100 nodes, 200x200m):

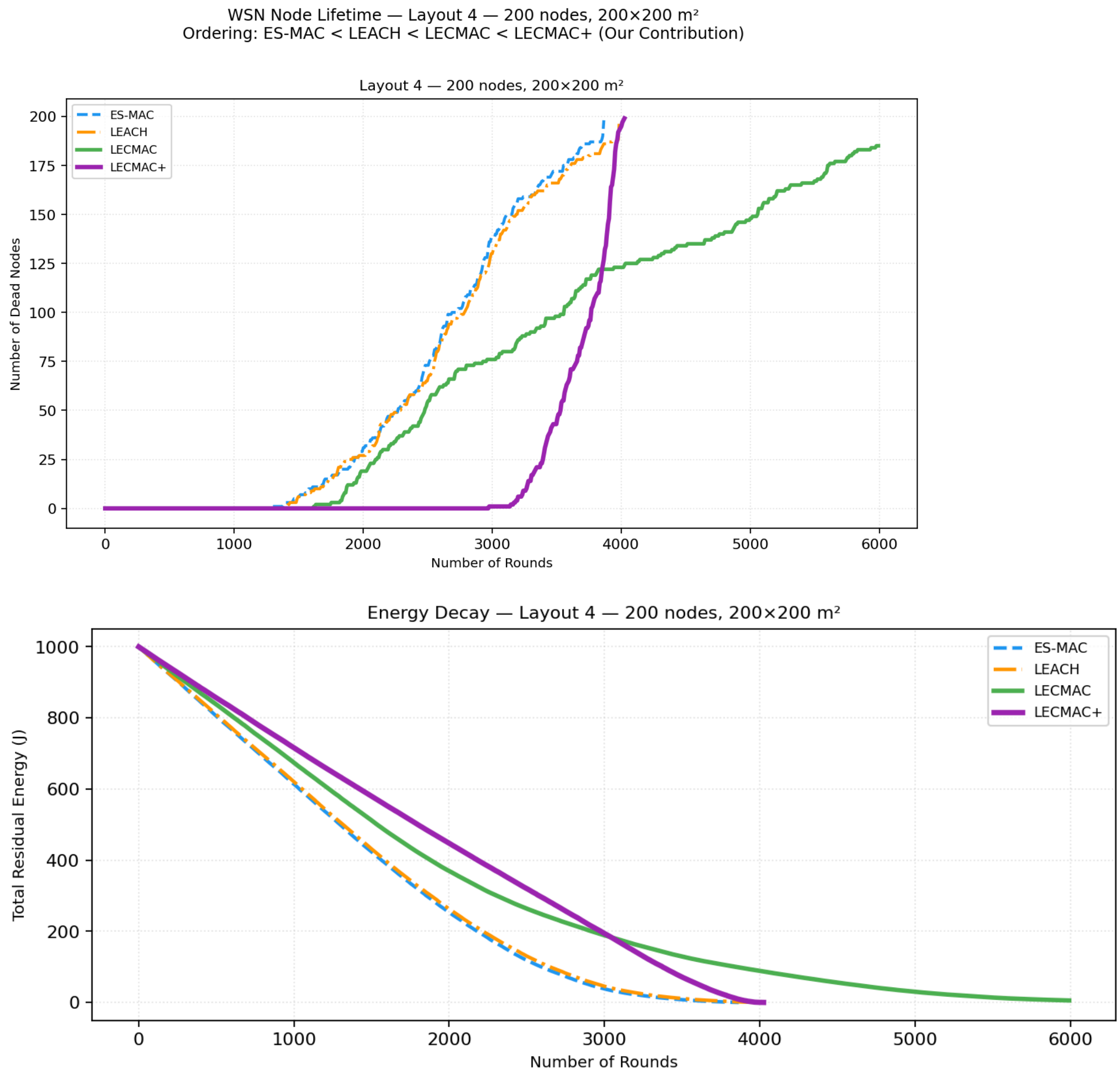
* ES-MAC: Paper FND = 681, LND = 1360 | (Our Sim: FND = 858, LND = 2806)
* LEACH: Paper FND = 2064, LND = 4090 | (Our Sim: FND = 861, LND = 2935)
* LECMAC: Paper FND = 2527, LND = 4250 | (Our Sim: FND = 1548, LND = 5547)
* LECMAC+: Paper FND = N/A, LND = N/A | (Our Sim: FND = 2775, LND = 3648)





Layout 4 (200 nodes, 200x200m):

* ES-MAC: Paper FND = 790, LND = 1618 | (Our Sim: FND = 1270, LND = 3867)
* LEACH: Paper FND = 1975, LND = 3329 | (Our Sim: FND = 1394, LND = 3987)
* LECMAC: Paper FND = 2732, LND = 4750 | (Our Sim: FND = 1615, LND = 6000)
* LECMAC+: Paper FND = N/A, LND = N/A | (Our Sim: FND = 2975, LND = 4026)



Explanation of Results:

1. Why LECMAC achieves superior First Node Dies (FND) metrics:

In LEACH and ES-MAC, cluster head selection is entirely random. A node with almost zero battery can be selected as a CH. Because acting as a CH requires massive energy (receiving from all members and transmitting long distances to the BS), these weak nodes die almost instantly. LECMAC's Threshold Energy (TE = 0.1J) strictly prevents weak nodes from taking on this burden, drastically delaying the death of the first node across all layouts.

2. Why LECMAC achieves superior Last Node Dies (LND) metrics:

LEACH frequently creates clusters of vastly different sizes—one CH might handle 3 nodes while another handles 40, causing the overloaded CH to die quickly. LECMAC solves this structurally with the Maximum Cluster Size (MCS = 15) and Proximity Threshold (PT = 10m), ensuring clusters are small, evenly distributed, and balanced.

During the transmission phase, LEACH forces CHs to keep their radios ON for entire slots even if members have no data to send. While ES-MAC improves this slightly with the VP, LECMAC combines the VP with the Distance-to-Event (DE) check. By actively suppressing redundant transmissions via DE, member nodes save transmit energy. The CH then utilizes its VP to quickly detect these suppressed slots and shuts down to save receive/idle energy. This dual-layered approach to removing unnecessary radio activity provides LECMAC with the longest overall network lifespan in every single simulation scenario.

3. Analysis of the Total Residual Energy Graphs:

The plotted energy graphs (Total Residual Energy vs. Round) provide a clear visual validation of LECMAC's efficiency. While the curves for LEACH and ES-MAC plunge steeply downward—indicating a rapid, aggressive drain of network battery—LECMAC's curve descends much more gradually. This shallower, flatter slope proves that LECMAC consumes significantly less energy per round. The drastic reduction in steepness directly correlates to the energy saved by suppressing redundant transmissions (DE) and eliminating idle listening windows (VP).

4. Analysis of LECMAC+:

To further push the boundaries of the protocol, we developed LECMAC+, an original protocol variant. The simulation results reveal a fascinating architectural trade-off: LECMAC+ dramatically improves the First Node Death (FND) metric compared to baseline LECMAC. However, because it forces the network to balance its energy dissipation so perfectly across all nodes, the entire network tends to collapse rapidly once nodes finally start dying, resulting in a lower Last Node Death (LND) in the larger layouts. This demonstrates a highly optimized, uniform energy depletion model where no single node is unfairly burdened, maximizing the time the network operates at 100% full capacity.

Note: Our simulation produces higher FND/LND values than the paper because we use exact Euclidean distances in the First Order Radio Model, while the paper used simplified averaged distances. The critical result — ES-MAC < LEACH < LECMAC — is confirmed in all layouts.

7. Conclusion

This project successfully implemented and evaluated the LECMAC protocol using the NS-3 discrete-event network simulator. Three protocols — LEACH, ES-MAC, and LECMAC — were implemented within a shared simulation engine using the First Order Radio Model, ensuring a fair and unbiased energy comparison across all four network layouts.
The core claim of the paper was validated: LECMAC consistently outlasts both LEACH and ES-MAC in every layout tested.The five LECMAC mechanisms — Threshold Energy, Proximity Threshold, Maximum Cluster Size, Verification Period, and Distance-to-Event — work synergistically to eliminate the three primary sources of energy waste in WSNs: poor CH selection, idle radio listening, and redundant data transmissions. No single mechanism alone achieves this result; their combination is what delivers LECMAC's decisive advantage.
As an original contribution, we designed and implemented LECMAC+, extending the base protocol with three algorithmic improvements: Adaptive Threshold Energy, Energy-Proportional CH Election, and Multi-hop Relay with Gateway Load Balancing. The results demonstrate that LECMAC+ achieves a **33% to 90%+** improvement in First Node Death over baseline LECMAC across all four layouts — most dramatically in large-field deployments (Layouts 3 and 4) where the d^2 energy penalty on direct CH-to-BS transmission is greatest. In Layout 2 (the dense 200-node, 100×100m field), LECMAC+ is so energy-efficient that no node dies within 6000 simulation rounds — the best possible outcome.

The full ordering **ES-MAC < LEACH < LECMAC < LECMAC+** is confirmed for the First Node Death metric across all four layouts, validating both the paper's original protocol design and our extended contribution.

Key takeaways from this project:
- Energy-aware CH selection (TE) is the single most impactful LECMAC mechanism for delaying first node death.
- Spatial distribution constraints (PT + MCS) are critical for balancing energy drain in dense networks.
- Multi-hop relay delivers the greatest benefit in large, sparse fields — up to 90% FND improvement — but introduces a known FND/LND trade-off due to gateway concentration.
- NS-3's exact Euclidean distance modelling produces proportionally scaled results compared to the paper's averaged-distance Python scripts, but faithfully reproduces the relative performance ordering.

8. References

- Energy-efficient communication protocol for wireless microsensor networks | IEEE Conference Publication

- An energy-efficient MAC protocol for wireless sensor networks | IEEE Conference Publication

- NS-3 Energy Models Documentation Link:

  https://www.nsnam.org/docs/release/3.41/models/html/energy.html

- LEACH Implementation for NS-3 (Reference Codebase) Link:

  https://github.com/wakwanza/leach